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Cleaning device for removing allergen particles from hair

Abstract

A cleaning device (**10, 100, 200**) for removing pollen and/or allergen particles from human and allergen particles from human and animal hair, comprising a handle device handle device (**12**) and a particle pick-up device (**14, 214**) detachably connected thereto (**14, 214**), wherein the particle pick-up device (**14, 214**) comprises a removal device (**20**) for removing the pollen and/or allergen particles. (FIG. **8**) Translated with DeepL.com (free version)

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Background/Summary

RELATED APPLICATIONS [0001] The application refers to the provisional application No. 63/561,745

DESCRIPTION
[0002] The invention relates to a cleaning device for removing allergen particles, in particular for removing allergen particles from human or animal hair.
[0003] Allergen particles are a serious burden for allergy sufferers. In particular, people who are allergic to certain pollens are seasonally burdened and impaired by allergic reactions as a result of contact with allergen particles, for example with various flower pollens. Various methods and medicines are known to alleviate or combat the allergic reaction and/or its symptoms.
[0004] Antihistamines are available in pharmacies and can be prescribed by specialists and general practitioners. This is done in particular for severe symptoms. Most of the active ingredients used as antihistamines increase tiredness and make you sleepy. Furthermore, these substances do not act against a specific allergy, but block a mechanism of the symptoms that occur by blocking the binding sites of histamine.
[0005] Another well-known treatment method is hyposensitization. In this procedure, which can be used for immunoglobulin E-mediated type I allergies, allergen extracts are administered to the human body at different intervals over a period of several years. In the long term, this leads to habituation to the respective allergen, so that the allergic reaction is ultimately absent or at least attenuated. This procedure is very time-consuming, as the person being treated is administered the allergen extract in the form of injections on a weekly basis at the beginning and later at monthly intervals. Each of these treatments is again time-consuming, as the treated person has to wait a certain period of time, usually 30 minutes, with the treating doctor after administration of the allergen extracts, as anaphylactic shock could possibly occur during this time.
[0006] Eye drops and nasal sprays are known to relieve common symptoms such as redness and itching of the eyes, runny nose or frequent sneezing, as well as asthmatic effects on the lungs, eye drops and nasal sprays, for example, are known. Cortisone is often used in these preparations, so they should not be taken over a long period of time or permanently if possible.
[0007] Homeopathic treatment methods and products are also available, for example acupuncture or black cumin oil, which are also intended to alleviate symptoms. However, there is no scientific evidence that these have any effect.
[0008] Another challenge is that each person treated reacts differently to treatment forms and remedies, meaning that success can vary greatly from person to person. Many people become drowsy from antihistamines, but the allergic reaction is also greatly reduced. Other people remain awake, but do not feel much effect from the symptomatic treatment.
[0009] For pollen allergy sufferers, the strength of the pollen count is relevant on a daily basis, which is usually stronger during the day than at night. Pollen particles therefore reach and enter the human body, especially the mucous membranes, where they trigger allergic reactions, if you are outside your home during the day or at night.
[0010] Pollen allergy sufferers with scalp hair have the particular problem that pollen particles can get caught in their hair and be carried into the home. Clothing with many fibers can also bring pollen into the home. If the allergy sufferer does not wash their hair immediately after arriving home, the pollen particles can get into the rooms of the home and also into furniture. The allergic substances in the pollen particles can persist in the pollen particles for several months and trigger allergic reactions even after this period.
[0011] It is therefore generally recommended that pollen allergy sufferers do not change in their bedroom, but instead change their clothes in the bathroom or in rooms where they do not sleep.

This limits the amount of pollen that enters the bedroom in particular. The aim is to prevent allergic reactions from interrupting the night-time rest phase in particular. It is also recommended to wash your hair, face and neck immediately to thoroughly remove the pollen. However, repeated washing of the hair can lead to the hair structure being destroyed. People therefore want to avoid washing their hair too often as far as possible. Even with repeated leaving the house during the course of the day, constant hair washing would impair both health and a regular daily routine.
[0012] U.S. Pat. No. 4,013,086 A discloses an essentially symmetrically foldable comb housing into which a flat filter material can be clamped.
[0013] ITMI 10 2011 901 974 697 discloses a comb with an impregnable felt arranged in the area of the comb teeth for applying care liquids to hair, whereby the comb can have a liquid container for holding the care liquid.
[0014] DE 20 2012 000 354 U1 describes a hairbrush for picking up dirt.
[0015] EP 3 838 065 A1 describes a hairbrush with bristles that can be extended by means of a rotary knob.
[0016] Against this background, the invention has the task of providing an easy-to-use device for removing allergen particles from hair which avoids the above-mentioned disadvantages.
[0017] The task is solved by a cleaning device according to claim **1**. Further advantageous embodiments are the subject of the dependent claims.
[0018] To solve the problem, a cleaning device for removing pollen and/or allergen particles from human and animal hair is proposed, which has a handle device and a particle pick-up device detachably connected thereto, wherein the particle pick-up device has a removal device for removing the pollen and/or allergen particles from the hair.
[0019] Such a device allows an allergy sufferer to easily and quickly remove pollen that gets into the scalp hair during everyday use. Cleaning and cleansing the hair in this way helps prevent allergic reactions in the home and, if possible, prevents them altogether. This can also reduce the need for further treatment, for example with antihistamines.
Specification
[0020] In some embodiments, the cleansing device may be designed as a brush comprising at least one housing part, which comprises a plurality of bristles, wherein the particle pick-up device has a plurality of openings for the bristles to pass through, whereby the particle pick-up device can be attached to the bristles.
[0021] The fact that the bristles protrude through the openings of the particle pick-up device ensures good contact between the hairs and the particle pick-up device.
[0022] In some embodiments, one of the housing parts has an at least partially circumferential and protruding edge, which has a mechanical latching or clamping device for detachable attachment of the particle holding device.
[0023] This makes it particularly easy to mount the particle pick-up device on the housing part. It is therefore particularly easy for the user to change the particle holder. In particular, no tools are required for this.
[0024] In some embodiments, an engagement recess is provided in an area of the edge, into which a tool and/or a finger can be inserted to remove the particle pick-up device from the mechanical latching or clamping device.
[0025] This makes it even easier for the user to change the particle holding device.
[0026] In some embodiments, the bristles are arranged on a movable bristle holder, wherein the bristle holder is movable within the housing parts such that the bristles can be retracted and extended from the housing part, wherein the bristles can be retracted into the housing part such that they do not or only slightly protrude from the particle pick-up device and/or the housing part.
[0027] As a result, the bristles can be moved completely or almost completely into the housing part. As a result, such a brush requires less space during transportation and the bristles are protected from external influences.

[0028] In some embodiments, the brush has a rotating section coupled by means of a linear converter to a guide carriage having guide openings for receiving a section of the bristle holder, in particular a guide pin, wherein the mediation of the linear converter and the guide carriage is arranged in such a way that the guide carriage is coupled to the guide carriage is designed in such a way that it converts a rotary movement of the rotary section into a linear movement of the bristle holder and thus of the bristles out of or into the housing part.
[0029] This makes the brush particularly precise to operate. By turning the rotating section, the bristles can be pushed out of the brush exactly as far as desired. Furthermore, there is no need for any externally accessible sliding mechanisms that could get caught in clothing or other objects during transportation, for example, and thus be activated.
[0030] In some embodiments, the bristles can be detachably fixed in their extreme positions, i.e. in a fully extended state and in a fully retracted state.
[0031] This prevents the bristles from slipping out of the bristle guide openings when they are retracted into the housing, which improves the handling of the brush **100**. It is also possible for the outer surface to be flat when the bristles are retracted, as the bristles can be completely retracted into the housing without a safety margin.
[0032] In some embodiments, the distances between the centers of gravity of the bristles are 0.75 cm to 2.5 cm, 1 cm to 1.5 cm, 1.2 cm to 1.5 cm, 1 cm, 1.2 cm, 1.25 cm or 1.5 cm, for example.
[0033] This provides sufficient space between the bristles to maximize the contact surface of the pick-up device.
[0034] In some embodiments, the bristles are arranged in a grid or in lines offset from one another, wherein a row of a grid or a line comprises 2 to 8, 3 to 5 or 3 to 4 bristles, and wherein 10 to 15, 11 to 12 or 11 rows or lines are provided.
[0035] This partially compensates for the increased adhesion of the removal device, which makes it more difficult to pass the brush through the hair, and at least does not make it more difficult to use the brush due to a large number of bristles.
[0036] In some embodiments, the cleaning device is designed as a comb, wherein the handle device is a handle part and the particle pick-up device is a omb part having prongs, wherein the comb part has at least one recess as a receptacle for a removal device.
[0037] The fact that the removal device is arranged in a recess makes it more difficult for the removal device to be displaced or pulled off during use, for example by adhesion to hair.
[0038] In some embodiments, at least one receptacle extends over a side surface of the comb part, wherein the side surface is delimited by a base of prongs.
[0039] As a result, the prongs themselves are free and serve to guide the hair along the comb.
[0040] In some embodiments, at least one receptacle is arranged in a contact surface arranged between two prongs.
[0041] As the hair is passed between the tines, the area between the tines is particularly exposed to the hair. A particle pick-up device that is inserted into such a holder can be used particularly efficiently at this point.
[0042] In some embodiments, a distance between two tines is more than 12 mm, more than 15 mm or more than 17 mm and/or the comb part has three, four, five, six, seven, eight, nine, ten, eleven or twelve tines and/or the tines have a conical or pyramid-like basic shape.
[0043] These embodiments ensure that the hair is cleaned as efficiently as possible, as the tines designed in this way are particularly easy to guide through the hair. In addition, such embodiments have particularly large contact surfaces with which the hair can come into contact.

Description

[0044] Further features and variants of the invention can be seen in the attached figures, which merely show schematic embodiments of the invention. They show in detail:
[0045] FIG. **1** a side view of a cleaning device according to an embodiment of the present invention;
[0046] FIG. **2** an isometric view of the cleaning device of FIG. **1**;
[0047] FIG. **3** a front view of the cleaning device from FIG. **1**;
[0048] FIG. **4** a side view of a cleaning device according to a further embodiment of the present invention;
[0049] FIG. **5** an isometric view of the cleaning device of FIG. **4**;
[0050] FIG. **6** a front view of the cleaning device of FIG. **4**;
[0051] FIG. **7** an exploded view of a cleaning device configured as a brush according to an embodiment of the present invention;
[0052] FIG. **8** an isometric view of a cleaning device according to FIG. **7** with extended bristles;
[0053] FIG. **9** a longitudinal section through a cleaning device as shown in FIG. **7** with the bristles retracted;
[0054] FIG. **10** a longitudinal section through a cleaning device as shown in FIG. **7** with the bristles extended;
[0055] FIG. **11** a side view of a cleaning device configured as a comb according to an embodiment of the present inve
[0056] FIG. **12** a sectional view of the cleaning device according to FIG. **11**;
[0057] FIG. **13** an exploded view of the cleaning device according to FIG. **11** with removal devices;
[0058] FIG. **14** an isometric view of the handle part of a cleaning device according to FIG. **11**;
[0059] FIG. **15** an isometric view of the comb part of a cleaning device according to FIG. **11** an
[0060] FIG. **16** a 3D view of an underside of the handle part of FIG. **14**.
[0061] Some of the figures contain simplified, schematic representations. In some cases, identical reference symbols are used for identical or functionally identical, but possibly not identical, elements. Different views of the same elements may be scaled differently. Directional indications such as “left”, “right”, “top” and “bottom” are to be understood with reference to the respective figure and may vary in the individual representations with respect to the object shown.
[0062] Furthermore, in the description and the patent claims, the phrase “at least partially” includes the term “completely”. The formulation “one” includes the formulation “at least one”, unless explicitly stated otherwise.
[0063] A cleaning device **10** shown in FIG. **1** has a handle device **12** and a particle pick-up device **14**. The particle pick-up device **14** has 6 engaging devices **16**, which project parallel to each other from a base section **18**. The engaging devices **16** are arranged in a row along a straight line. The particle pick-up device **14** is covered by a pick-up device **20**, which has a retaining layer of microfiber fabric. The microfiber fabric has the property of attracting allergen particles, for example pollen particles, so that they adhere or stick to the retention layer.
[0064] A feed-through section **22** is provided between each of the gripping devices **16**. The feed-through section **22** is designed in such a way that hair that is fed through it comes into contact with the removal device **20** and thus the retention layer over as large an area as possible.
[0065] The cleaning device **10** thus forms a kind of comb by means of which allergen particles can be removed from hair. For this purpose, the cleaning device **10** is passed through the hair several times just like a comb. In this way, the retention layer of the removal device **20**, which covers at least part of the surfaces of the engaging devices **16**, for example the surfaces of the teeth of the comb, comes into contact with the surface of as many hairs as possible. The allergen particles are thereby removed from the hair by the retaining layer of the removal device **20** and retained in the retaining layer.

[0066] When the removal device **20** has picked up a certain number of allergen particles, its efficiency is reduced. Depending on the material used for the retention layer, the pick-up device **20** can then be cleaned. If the material cannot be cleaned, the particle pick-up device **14** must be replaced with a new particle pick-up device **14**.
[0067] In some embodiments, it is possible for the removal device **20** to be detachably attached to the particle pick-up device **14**, for example, so that only the removal device **20** needs to be replaced.
[0068] The cleaning device **10** is suitable for removing pollen particles from human hair, synthetic hair and animal hair. The handle device **12** serves to hold the cleaning device **10** and to be able to move and guide it for removing pollen particles.
[0069] The particle pick-up device **14** can be replaced as soon as a filter material used in the pick-up device **20** has reached a maximum pollen particle pick-up capacity. The engaging devices **16** of the particle pick-up device **14** can form prongs, teeth, teeth or rods, which are guided through the hair in order to simultaneously touch as many hairs as possible and to brush or comb the pollen particles adhering to the hair into the filter material of the pick-up device **20** by means of a combing movement.
[0070] In order to enable the particle pick-up device **14** to be replaced, the handle device **12** and the particle pick-up device **14**, as shown in FIG. **2**, have first and second fastening sections **24**, **26** which can be inserted into one another. The first fastening section **24** has a substantially circular cylindrical shape. The second fastening section **26** has a recess into which the first fastening section **24** can be inserted along an axis of the circular cylindrical shape. In this embodiment, the second fastening section **26** of the handle device **12** thus embraces the first fastening section **24** of the particle pick-up device **14**.
[0071] In addition, the fastening section **26** has a groove **28** at which the engaging devices **16** can pass through the second fastening section **26**.
[0072] To replace the particle pick-up device **14**, the particle pick-up device **14** is pulled out of the second fastening section **26** along the axis and a new particle pick-up device **14** is then pushed into the second fastening section **26**.
[0073] As can be seen in the section in FIG. **3**, the particle holder **14** has a base structure **29**, which gives the particle holder **14** stability. The removal device **20** is arranged on surfaces of the base structure **29**. As a result, the pick-up device **20** does not require its own stability and can be constructed more simply.
[0074] In further embodiments, one of which is shown in FIGS. **4** to **6**, the particle pick-up device **14** has a first side **30** on which the engaging devices **16** are distributed. In these embodiments, the cleaning device **10** thus forms a brush, in particular a hairbrush. The removal device **20** is designed in such a way that the engaging devices **16** penetrate it. For this purpose, the removal device **20** has openings. This means that the removal device **20** can be attached to the engaging devices **16**.
[0075] The particle pick-up device **14** can be detachably fastened in the handle device **12**. For example, a push button or other actuating device can be provided to release the particle pick-up device **14**.
[0076] The removal device **20** is merely clamped to the gripping devices **16** for temporary attachment, for example. In some embodiments, for example, a simple detachable connection, such as a clamp or clip connection, may be provided for fastening the removal device **20**.
[0077] In some embodiments, the removal device **20** comprises, for example, a microfiber fabric, a microfiber woven fabric, a microfiber nonwoven fabric or generally a nonwoven fabric or a woven fabric. Microfiber nonwoven fabric is particularly well suited for use in the removal device **20**, as it has many densely packed fibers in the micrometer range to which allergen particles can adhere. In addition, microfiber nonwovens can be designed in such a way that they have an electrostatic charge, which attracts the allergen particles. This material is not known to have any properties that are hazardous to human health.

[0078] The invention is not only suitable for reliably removing allergen particles from human hair, but also from the hair of animals after a walk, for example. This can prevent allergen particles, in particular pollen, from spreading in a home. This reduces the need to treat symptoms with medication. In addition, the cleaning device **10** is portable and compactly stowable so that it can be taken to different places very easily.
[0079] In some embodiments, the cleaning device **10** can be designed as a brush **100**, as shown as an example in FIGS. **7** to **10**. The brush **100** has a housing formed from a lower housing part **102**, which has the handle section **12**, and an upper housing part **104**, and a bristle holder **106**, on which the engaging devices **16** designed as bristles **108** are arranged.
[0080] A base surface of the bristle holder **106** is designed such that it has guide sections **110** at its edges. The guide sections **110** are shaped in such a way that they can rest with lateral surfaces against walls of the housing parts **102**, **104** for guidance. As a result, these guide sections **110** can, for example, serve to arrange and guide the bristle holder **106** in the housing parts **102**, **104**. In this embodiment example, the bristle holder essentially has a rectangle with rounded corners as a base surface, wherein recesses are arranged on the sides and the guide sections **110** are formed by the rounded corners. The housing part **104** can furthermore have guide rails which run along a displacement direction of the bristles **108** through the bristle guide openings **118**, so that the individual bristles **108** are permanently centered with respect to the respective bristle guide openings **118** and/or holes in the housing part **104** and cannot slip with respect thereto.
[0081] The bristles **108** are aligned with bristle guide openings **118** disposed in the upper housing portion **104** so that, once the bristles **108** are disposed in the bristle guide openings **118**, they are restricted in their movement such that they can substantially only move back and forth through the bristle guide openings **118**. The bristles **108** and thus also the bristle holder **106** connected to them are thus also guided by means of the bristle guide openings **118**.
[0082] The bristles **108** can be designed such that, in the retracted state, they are aligned with an outer surface, for example an outer surface of the particle receiving device **14**, thereby forming, for example, a substantially flat surface.
[0083] The bristles **108** can be flexible or rigid. For example, the bristles **108** may have a conical shape.
[0084] Compared to a conventional hairbrush, the number of bristles **108** is reduced so that the bristles **108** are larger and spaced further can be arranged at a distance from each other. This results in a large contact surface that is still comfortable to use. In addition, the usable area of the removal device **20** is increased.
[0085] In some embodiments, a number of the bristles **108** may be, for example, 25 to 50, 30 to 45, 35 to 40, or 39. In some embodiments, the bristles **108** are arranged on the bristle holder **106** for example in a grid or in lines offset from each other, wherein a row of a grid or a line comprises for example 2 to 8, 3 to 5 or 3 to 4 bristles **108**, wherein for example 10 to 15, 11 to 12 or 11 rows or lines may be provided, which may for example each comprise a different number of bristles **108**. In some embodiments, a distance between center points and/or center axes of the bristle guide openings **118** and thus also between centers of gravity and/or, in the case of rotationally symmetric bristles **108**, center axes of the bristles **108** may be, for example, 0.75 cm to 2.5 cm, 1 cm to 1.5 cm, 1.2 cm to 1.5 cm, 1 cm, 1.2 cm, 1.25 cm or 1.5 cm. In some embodiments, the bristles **108** are spaced apart from an outer edge of the bristle holder **106** by at least 6 mm or at least 8 mm, for example. It is apparent from these dimensions that the brush **100** is usually larger than conventional hairbrushes.
[0086] Guide pins **112** are arranged on long sides of the bristle holder **106**, which can be inserted into guide openings **114** of a guide carriage **116**. The guide carriage **116** has first guide rails **120** on a side facing away from the bristle holder **106**, which are arranged to engage in second guide rails **122** arranged in the lower housing part **104**. Due to this engagement of the guide rails **120**, **122** with one another, the guide carriage **116** is only displaceable in a control direction extending along

the guide rails **120**, **122**.

[0087] The guide carriage **116** has second guide pins **124** in an actuation area **128**, which engage in elongated holes **126** arranged in a grip area of the lower housing part **102**. The elongated holes **126** are arranged and aligned such that the guide carriage **116** is linearly displaceable along the control direction.

[0088] The bristle holder **106** cannot be displaced in the control direction, since it rests with its guide sections **110** in this direction against one of the housing parts **102**, **104**. At the same time, the guide pins **112** of the bristle holder **106** are inserted into the guide openings **114**. If the guide carriage **116** is displaced in the control direction, then the guide pins **112** follow the shape of the guide openings **114**. The guide openings **114** are designed, for example, as elongated holes which are arranged at an angle to the control direction.

[0089] In some embodiments, the guide openings **114** have, for example, a short retaining section at the end that is arranged parallel to the control direction. The retaining section can be designed in such a way that the guide pins **112** can engage in at least one of their end positions in such a way that an increased operating force is required to move them out of the end position. At the same time, this can have the effect that, for example, the bristle holder **106** is temporarily locked in the extended position of the bristles **108** for use.

[0090] In some embodiments, the rotating section **132** may be stabilized and/or fixed by a small trough in the rotating mechanism within the rotating section at the highest point of extension and retraction of the bristles **108** by a latching and/or clamping mechanism. This serves to prevent the bristles **108** from retracting again directly on contact with the hair and scalp or from accidentally extending out of the housing. A counter-movement simply releases the stabilization or fixation and allows the bristles **108** to move again.

[0091] Due to the elongated holes as well as the guidance through the housing parts **102**, **104**, the bristle holder **106** is moved with its bristles **108** through the bristle guide openings **118** perpendicular to the control direction when the guide carriage **116** is moved in the control direction. The bristle guide openings **118** and thus also the bristles **108** are arranged, for example, in rows and/or columns, wherein the rows and/or columns can be arranged offset relative to one another.

[0092] The bristles **108** can be moved into the housing by means of the guide carriage **116** in such a way that they do not protrude from the particle receiving device **14** and/or the openings **118** of the upper housing part **104**. Bristle guides **148** can be provided on an inner side of the housing part **104**, which hold the bristles **108** in position even when they are retracted into the housing. This prevents the bristles **108** from slipping out of the bristle guide openings **118** when they are retracted into the housing, which improves the handling of the brush **100**. Thus, it is also possible for the outer surface to be flat when the bristles **108** are retracted, since the bristles **108** can be completely retracted into the housing without a safety margin. In some embodiments, the bristle guides **148** are designed as short, hollow cylinder-like projections arranged around an inner edge of the bristle guide openings **118**.

[0093] A rotary section **132** is provided for operating the guide carriage, which is terminated by means of an end section **130**. A rotary movement of the rotary section **132** carried out by an operator is converted into a movement in the control direction by means of a linear converter integrated therein. By rotating the rotating section **132**, the bristles **108** can thus be extended from the upper housing part **104** and retracted into it completely, for example.

[0094] The housing parts **102**, **104** can have, for example, latching lugs **134** and/or latching openings **136** which latch into one another when the housing parts **102**, **104** are assembled and thus provide sufficient load-bearing capacity.

[0095] The particle pick-up device **14** has a carrier **140** in addition to the pick-up device **20**. The removal device **20** is arranged and attached to the carrier **140**, so that the carrier **140** stabilizes and supports the removal device **20**. The upper housing part **104** has a recess **142** with an edge **138**, which at least partially has a clamping or latching projection, by means of which the carrier **140**

and/or the particle pick-up device **14** can be detachably fastened in the recess **142**. The rim **138** may, for example, be 1 mm to 3 mm, 2 mm to 5 mm or 2 mm high. In some embodiments, the height of the rim **138** may be adapted to the thickness of the carrier **140** and/or the particle receiving device **14**. In order to simplify the assembly of the brush **100**, the carrier **140** can be designed without a removal device **20** in an edge region that can be clamped by the edge **138**.
[0096] The carrier **140** can, for example, be formed from a deformation-resistant material and has a thickness of, for example, 1 mm to 5 mm, 2 mm to 3 mm or 2 mm.
[0097] In particular, the take-off device **20** has a nonwoven material or is formed by such a material. The nonwoven material should be selected in such a way that it allows particularly good removal of pollen. Nonwoven material made of PP+PAN fibers has proven to be a suitable material for this purpose.
[0098] In some embodiments, the removal device **20** is firmly connected to the carrier **140**, for example with a material bond. To accommodate the removal device **20** and/or to simplify its alignment on the carrier **140**, the carrier **140** may have a recess whose depth is adapted to the thickness of the removal device **20**, for example. In some embodiments, a depth of the recess may correspond to a thickness of the take-off device **20** so as to provide a flat surface, or the material of the take-off device **20** may protrude slightly.
[0099] The upper housing portion **104** further includes an engagement recess **144** into which a user may engage, for example, with a finger, fingertip, or suitable narrow object to disengage the particle pickup device **14** from the recess **142**. The particle receiving device **14** may also have an engagement portion, for example in the region of the engagement recess **144**, to facilitate its removal from the recess **142**.
[0100] The particle receiving device **14** has openings **146** in the support **140** and the removal device **20** which, when the particle receiving device **14** is disposed in the recess **142**, are aligned with the bristle guide openings **118** so that the bristles **108** can be pushed through the upper housing portion **104** and the particle receiving device **14**.
[0101] In some embodiments, the cleaning device **10** may be formed as a comb **200** as shown as an example in FIGS. **11** to **16**. The comb **200** has a handle part **212** as the handle device **12** and an exchangeable comb part **214** as the particle pick-up device **14**. The comb part **214** has a base section **218**, from which a plurality of prongs **216** protrude. A receptacle **224** is provided on at least one side of the base section **218**, into which a pick-up device **20** can be inserted. The prongs **216** of the comb portion **214** form engagement devices **16**, between which spacings **222** are formed as lead-through sections **22**. A contact surface **226** is provided between each of the prongs **216** as a transition, against which hair is pulled during combing.
[0102] The receptacles **224** are arranged, for example, on both sides of the base section **218**. In addition, a receptacle **224** is arranged between the prongs **216** in at least one of the contact surfaces **226**.
[0103] The receptacles **224** are designed, for example, as recesses and each have a base surface and/or side surfaces to which the removal devices **20** can be attached. This ensures, in particular, that the removal devices **20** are each securely fastened to the comb part **214** and cannot be displaced or pulled out of the receptacles **224** by adhesion to the hairs, for example on contact with the hairs. The pick-off devices **20** can be non-detachably connected to sections of the receptacles **224**, for example by adhesive bonding. Since the receptacles **224** are designed as recesses which are, for example, between 1.5 and 3 mm, between 1.5 and 2.5 mm or between 1.8 and 2.2 mm deep, they can have at least partially circumferential support edges on which the removal device **20** can be supported in each case.
[0104] In some embodiments, for example, the base portion **218** extends over between 50% and 90% of a height h of the comb portion **214**, such that the prongs **216** extend over between 20% and 50% of the height of the comb portion **214**. Prongs **216** may, for example, have a spherical or conical shape at least in sections or the shape of an at least three-sided pyramid or a four-sided

pyramid.
[0105] In contrast to conventional combs, which are mainly used for separating strands of hair and for combing out knots, it can be advantageous for comfortable use of the comb **200** if there is a balanced relationship between the combing effect, i.e. the order and/or arrangement of the hair, and the cleaning effect, for example in the form of pollen removal. In addition, it can be advantageous to find a balance between a combing effect and the cleaning effect in the form of pollen pick-up. Removal devices **20** are arranged in the holders **224**, which may, for example, have a fabric, in particular a microfiber fabric, which has an increased static friction when in contact with hair relative to a conventional comb tine with a smooth surface. In order to achieve a balanced ratio between combing effect and cleaning effect, in some embodiments it is provided, for example, that tips of the tines **216** have a spacing of between 1 cm and 2.5 cm, between 1.2 cm and 2.2 cm, between 1.5 cm and 1.9 cm or between 1.6 cm and 1.8 cm. As a result, in some embodiments, a comb portion **214** has, for example, only six to eight, three, four, five, six, seven, eight, nine, ten, eleven or twelve prongs **216**. Such embodiments can also provide particularly large contact surfaces with correspondingly large pick-up devices **20**, which make the cleaning process more convenient and efficient, as indicated above.
[0106] FIGS. **14** to **16** show details of the detachable connection between the handle part **212** and the comb part **214**. The handle part **212** has a guide opening **230** delimited by two guide strips **228**, into which a back section **236** of the comb part **214** can be inserted. The back section **236** has grooves **28** into which the guide strips **228** engage during insertion. The guide strips **228** and the grooves **28** form a positive connection when the comb **200** is assembled.
[0107] A detachable fastening device, by means of which the back section **236** can be detachably held in the guide opening **230**, can for example be formed by means of a latching or clamping connection. In some embodiments, for example, the comb portion **214** has latching clips **234** at the end, which can engage in latching grooves **232** of the handle portion **212** to form the releasable latching connection.
[0108] In some embodiments, particularly in embodiments configured as a brush **200**, the replaceable particle pickup device **14** is prepared for commercial exploitation such that the carrier **140** and the pickup device **20** are non-detachably connected to each other. In embodiments which are designed as a comb **200**, this preparation takes place by inserting the removal devices **20**, for example the nonwovens, into the receptacles **224**. The particle pick-up device **14** prepared in this way can be used to replace a used particle pick-up device **14** that is no longer efficient due to limited pollen pick-up capacity.
[0109] Providing the particle pick-up device **14** in this way has the advantage over embodiments that provide the pick-up device **20** separately that the pick-up device **20** can be used in a sterile manner, for example in an automatic production machine. If this is carried out by a user, the user can contaminate the removal device **20**, for example by pressing it by hand, so that its useful life is shortened.
[0110] In some embodiments, the handle device and the particle pick-up device are detachably connected to each other.
[0111] This makes it easy to replace the particle pick-up device if, for example, it can no longer pick up any more allergen particles after prolonged use.
[0112] In some embodiments, the pick-up device is detachably attached to the particle pick-up device
[0113] Depending on the embodiment, this allows the removal device alone, which removes the allergen particles from the hair, to be replaced. This reduces material consumption in an environmentally friendly manner.
[0114] In some embodiments, the particle pick-up device has a plurality of engaging devices for engaging between a plurality of hairs, with the pick-up device being arranged on a surface of the engaging devices.

[0115] This forms a kind of comb that can be guided through the hairs. The hairs sweep along the surface of the engaging devices on which the removal device is arranged. The removal device then removes the pollen particles with their allergens from the hair, which is pulled over it. In this way, an allergic reaction can be reduced, avoided or prevented by simply combing or brushing the hair.
[0116] In some embodiments, the removal device is arranged such that the surface of the engaging devices is at least partially covered by the removal device.
[0117] This increases the surface area on which the hair comes into contact with the removal device.
[0118] In some embodiments, the engaging devices are aligned parallel to each other.
[0119] If a particle pick-up device designed in this way is pulled through hair, a particularly uniform cleaning of the hair is achieved.
[0120] In some embodiments, the engaging devices are arranged on a first side of the particle pick-up device at an engaging portion, wherein the pick-up device is attachable to at least a plurality of the engaging devices such that it is arranged adjacent to the engaging portion.
[0121] Such a particle pick-up device is used, for example, to form a hairbrush, depending on the shape of the first side. By attaching the pick-up device to the engaging devices, a particularly large surface area of the pick-up device is provided for contact with the hair.
[0122] In some embodiments, the engaging devices are lined up along a straight line on the particle pick-up device.
[0123] Such a cleaning device forms a comb, for example. It can be designed to be particularly space-saving, so that a pollen allergy sufferer can take it with them when traveling, for example.
[0124] In some embodiments, the removal device has a retention layer for retaining allergen particles.
[0125] Such retention layers have, for example, a surface to which the allergen particles adhere due to a chemical reaction, for example sticking, and/or a physical process, for example by hooking fine structures, and can thus be removed from hair. Advantageously, the retention effect of the retention layer is stronger against allergen particles than against hair.
[0126] In some embodiments, the retention layer comprises a nonwoven or a fabric, in particular comprising microfibers.
[0127] Nonwovens, woven fabrics and microfiber fabrics, in particular microfiber nonwoven fabrics, are fine-pored and yet have a high density, which gives them ideal filter properties. In addition, many such materials have electrostatic properties so that they can generate an electrostatic field in their environment, which attracts allergen particles. PP (polypropylene) and PAN (polyacrylonitrile) fibers are a particularly advantageous material to use.
LIST OF REFERENCE SYMBOLS
[0128] **10** Cleaning device [0129] **12** Gripping device [0130] **14** Particle pick-up device [0131] **16** Engaging device [0132] **18** Base section [0133] **20** Removal device [0134] **22** Feed-through section [0135] **24** First fastening section [0136] **26** Second fastening section [0137] **28** Groove [0138] **29** Base structure [0139] **30** First side (of particle pick-up device **14**) [0140] **100** Brush [0141] **102** Lower housing section [0142] **104** Upper housing section [0143] **106** Bristle holder [0144] **108** Bristles [0145] **110** Guide section [0146] **112** Guide pin [0147] **114** Guide opening [0148] **116** Guide carriage [0149] **118** Bristle guide opening [0150] **120** First guide rail [0151] **122** Second guide rail [0152] **124** Second guide pin [0153] **126** Oblong hole [0154] **128** Actuation section [0155] **130** End section [0156] **132** Turning section [0157] **134** Latching lug [0158] **136** Latching opening [0159] **138** Edge [0160] **140** Carrier [0161] **142** Recess [0162] **144** Recessed recess **146** Opening [0163] **148** Bristle guide **200** Comb [0164] **212** Handle part [0165] **214** Comb part [0166] **216** Tine [0167] **218** Base section [0168] **222** Distance [0169] **224** Mounting [0170] **226** Contact surface [0171] **228** Guide strip [0172] **230** Guide opening [0173] **232** Locking groove [0174] **234** Locking clamp [0175] **236** Back section

Claims

1. Cleaning device (**10**, **100**, **200**) for removing pollen and/or allergen particles from human and animal hair, comprising a handle means (**12**) and a particle receiving means (**14**, **214**) detachably connected thereto, wherein the particle receiving means (**14**, **214**) comprises a removal means (**20**) for removing the pollen and/or allergen particles.
2. Cleaning device according to claim 1, characterized in that the cleaning device is designed as a brush (**100**), having at least one housing part (**102**, **104**), which has a plurality of bristles (**108**), the particle pick-up device (**14**) having a plurality of openings for the passage of the bristles (**108**), as a result of which the particle pick-up device (**14**) can be plugged onto the bristles (**108**).
3. Cleaning device according to claim 2, characterized in that one of the housing parts (**102**, **104**) has an at least partially circumferential and projecting edge (**138**) which has a mechanical latching or clamping device for releasably fastening the particle pick-up device (**14**).
4. Cleaning device according to claim 3, characterized in that an engagement recess (**144**) is provided in a region of the edge (**138**), into which recess a tool and/or a finger can be inserted for removing the particle pick-up device from the mechanical latching or clamping device.
5. The cleaning device according to any one of claims 2 to 4, characterized in that the bristles (**108**) are arranged on a movable bristle holder (**106**), wherein the bristle holder (**106**) is thus movable within the housing parts (**102**, **104**), that the bristles (**108**) can be moved in and out of the housing part (**102**, **104**), wherein the bristles (**108**) can be moved into the housing part (**102**, **104**) in such a way that they do not or only slightly protrude from the particle receiving device (**14**) and/or the housing part (**102**, **104**).
6. Cleaning device according to claim 5, characterized in that the brush (**100**) has a rotary section (**132**) which is coupled by means of a linear converter to a guide carriage (**116**) which has guide openings (**114**) for receiving a section of the bristle holder (**106**), in particular a guide pin (**112**), wherein the coupling of the linear converter and the guide carriage (**116**) is designed such that it converts a rotary movement of the rotary section (**132**) into a linear movement of the bristle holder (**106**) and thus of the bristles (**108**) out of or into the housing part (**102**, **104**).
7. Cleaning device according to claim 6, characterized in that the bristles (**108**) can be releasably fixed in their extreme positions, i.e. in a completely extended state and in a completely retracted state.
8. Cleaning device according to one of claims 2 to 7, characterized in that distances from centres of gravity of the bristles (**108**) are, for example, 0.75 cm to 2.5 cm, 1 cm to 1.5 cm, 1.2 cm to 1.5 cm, 1 cm, 1.2 cm, 1.25 cm or 1.5 cm.
9. Cleaning device according to one of claims 2 to 8, characterized in that the bristles (**108**) are arranged in a grid or in lines offset from one another, wherein a row of a grid or a line comprises 2 to 8, 3 to 5 or 3 to 4 bristles (**108**), and wherein 10 to 15, 11 to 12 or 11 rows or lines are provided.
10. Cleaning device according to claim 1, characterized in that the cleaning device is a comb (**200**), wherein the handle means (**12**) comprises a handle part (**212**) and the particle pick-up means (**14**) comprises a comb part (**214**) having prongs (**216**), characterized in that the comb part (**214**) comprises at least one receptacle (**224**) for receiving the pick-up means (**20**).
11. Cleaning device according to claim 10, characterized in that at least one recess extends as a receptacle (**224**) over a side surface of the comb part (**214**), the side surface being delimited by a base portion of the tines (**216**).
12. Cleaning device according to claim 10 or 11, characterized in that at least one receptacle (**214**) is arranged in a contact surface (**226**) arranged between two tines (**216**).
13. Cleaning device according to one of claims 10 to 12, characterized in that the tines (**216**) have a length along the comb part (**214**) of more than 12 mm, more than 15 mm or more than 17 mm and/or the distance between two tines (**216**) at the level of their base section is more than 9 mm,

more than 10 mm or more than 12 mm.

14. Cleaning device according to one of claims 10 to 13, characterized in that the comb part (**214**) has five to eight, three, four, five, six, seven, eight, nine, ten, eleven or twelve tines (**216**) and/or in that the tines (**216**) have a conical or pyramid-like basic shape.